KALAM (கககக) — AGRICULTURAL INSIGHTS ATLAS

Ambasamudram & Cheranmahadevi Taluks, Tirunelveli · Multi-Year EO Analytics (2023–2026)

CROPPING INTENSITY

12.6% Double-Cropped Cropland

CANOPY VIGOUR

95.6% Normal / Vigorous Cropland

ANNUAL WATER DEMAND

348.51 Million m³ (MCM) Gross Requirement

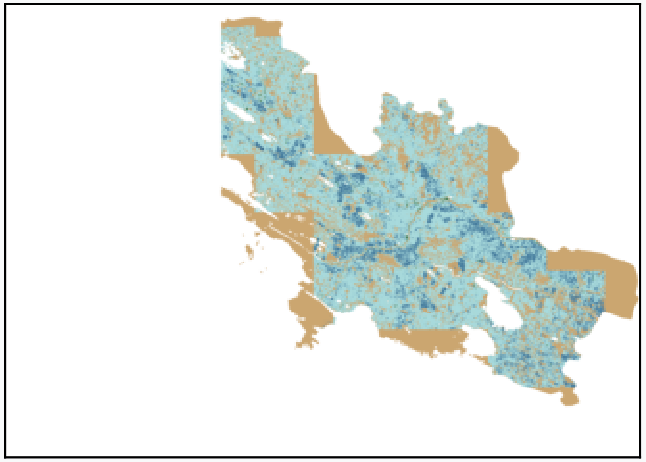
EXECUTIVE INSIGHT SUMMARY & POLICY ACTION ITEMS

- **Cropping Intensity:** 12.6% of cropland (5288.1 ha) in Ambasamudram & Cheranmahadevi is double-cropped with dual Kar and Pishanam cycles, alongside 0.9% under perennial banana canopy.
- **Transplanting Calendar:** Peak Kar transplanting occurs in May (46.6% of Kar fields), while main Pishanam sowing peaks in Nov (39.5% of Pishanam fields). 0.0% of Pishanam acreage experiences delayed transplanting beyond November.
- **Irrigation Dependency & Water Access:** 8.9% of paddy acreage is concentrated within 500 m of a river channel, canal or irrigation tank. Banana accounts for 0.8% of cropland in the immediate 0–250 m corridor, decaying sharply beyond 1 km.
- **Crop Change & Farmland Dynamics:** Evaluated 41,879.6 ha of 200,961.8 ha cropland (confidence ≥ 0.60): 26.3% of paddy and 0.8% of banana parcels remained stable across 2023–2026, with 705.9 ha (1.7%) identified as candidate paddy-to-banana diversification.
- **Crop Health & NDVI Anomaly:** 95.6% of analyzed cropland displays normal or above-normal vegetative vigour. Vegetative moisture/vigour stress is observed across 4.4% (1101.0 ha) of cultivated area.
- **Indicative Gross Irrigation Requirement:** Total indicative gross irrigation requirement across analyzed farmed plains is 348.5 MCM/year. Paddy cycles (Kar & Pishanam) account for 75.0% (261.4 MCM), while perennial banana requires 7.4 MCM (2.1%).

Kalam © 2026 · Problem Statement 1 · SustainAI / Mapathon · Contains modified Copernicus Sentinel data

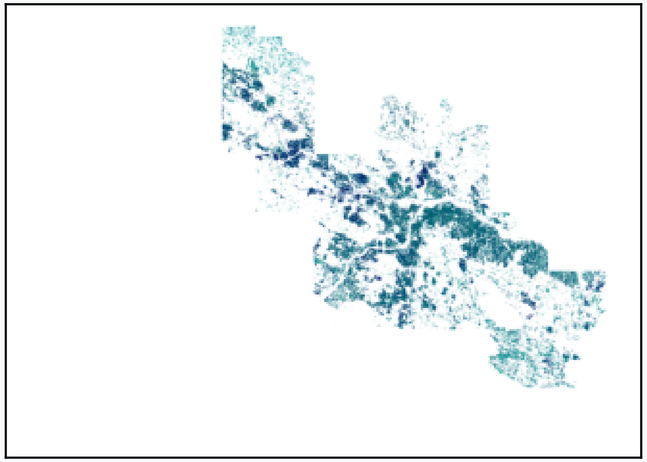
PHENOLOGY DYNAMICS & IRRIGATION ACCESS

01. Cropping Intensity



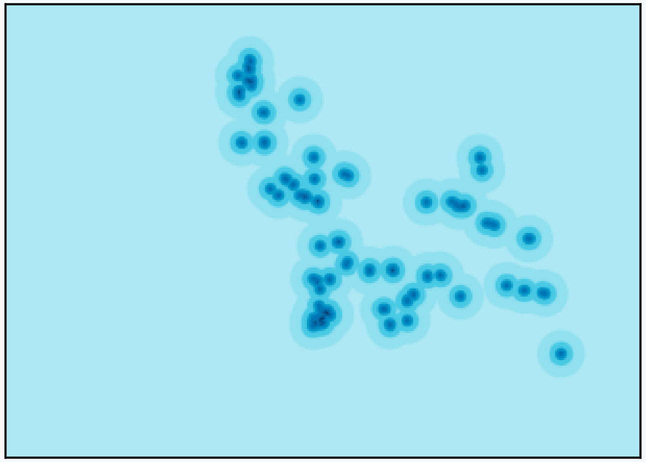
vated (7972.0 ha, 19.0%) | Single Crop (28259.8 ha, 67.5%) | Double Crop (5288.1 ha, 12.6%) | Triple Crop (2.3 ha, 0.0%)

02. Transplanting Calendar



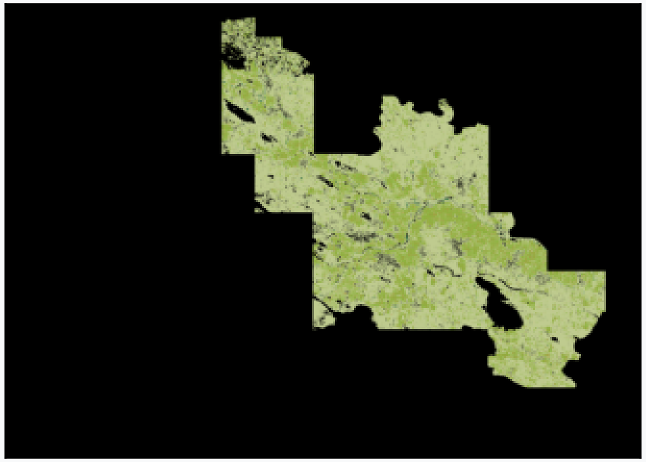
Kar Peak (Jun–Jul) | Kar Late (Aug) | Pishanam Peak (Oct) | Pishanam Late (Nov–Dec)

03. Irrigation Dependency



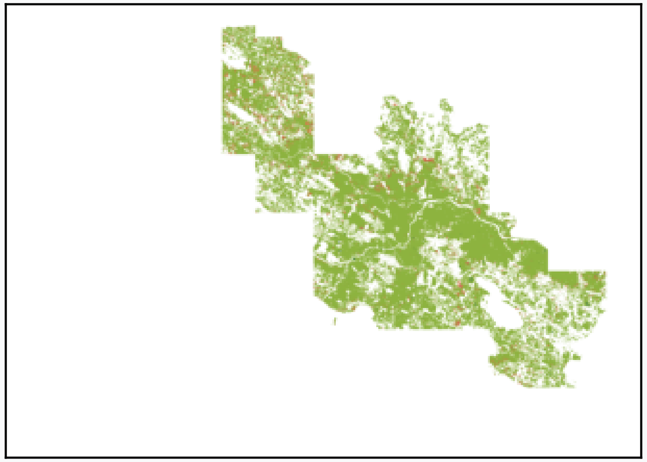
0 – 250 m | 250 – 500 m | 500 – 1,000 m | 1,000 – 2,000 m

FARMLAND DYNAMICS, VIGOUR & IRRIGATION DEMAND



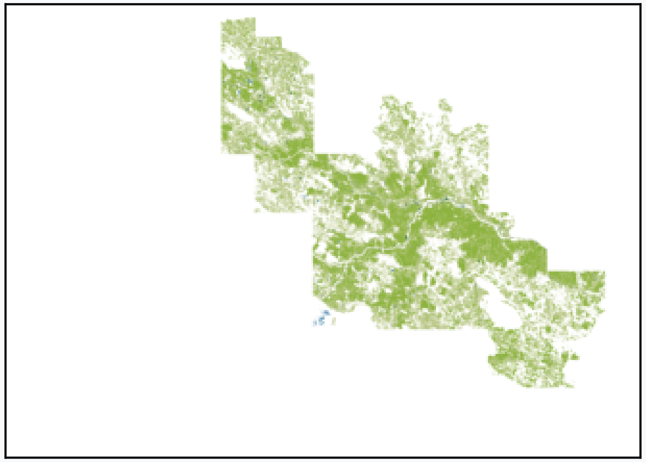
04. Crop Change & Loss

Stable Paddy | Stable Banana | Candidate Paddy → Banana | Banana → Paddy



05. NDVI Vigour Anomaly

Severe Stress (186.8 ha, 0.7%) | Stressed (914.2 ha, 3.7%) | Normal Vigour (23,815.3 ha, 95.2%) | Above Normal (93.0 ha, 0.4%)



06. Water Demand Budget

Kar Paddy (Short-duration) (117.76 MCM, 33.8%) | Pishanam Paddy (Main Monsoon) (143.61 MCM, 41.2%) | Perennial Banana (Year-round) (7.4 MCM, 2.1%